Electroweak baryogenesis is doubly forbidden in the Standard Model: a Lean-formalized two-pillar bridge to the chirality wall

John G. Roehm[1]

[1]*NetRxn Foundation*

(Dated: August 10, 2026)

We formalize the verdict that electroweak baryogenesis is forbidden in the Standard Model under *two independent* obstructions: *(i)* the chirality-wall obstruction at the gauging step, and *(ii)* the crossover nature of the SM electroweak phase transition. The first reflects the Wang $\mathbb{Z}_{16}$ anomaly classification [1]: the SM-no-ν_R fermion content gives $3 \times 15 = 45 \equiv 13 \not\equiv 0 \pmod{16}$, blocking non-perturbative lattice gauging [2] and depriving the sphaleron rate of a first-principles regulator. The second is a lattice result: the resummed-thermal-corrections endpoint at $m_H = 72.4 \pm 1.7$ GeV (Csikor, Fodor, and Heitger [4], refining Kajantie–Laine–Rummukainen–Shaposhnikov [3]) sits well below the PDG value $m_H = 125.20$ GeV [10]; the SM electroweak transition is a crossover, blocking sphaleron decoupling (Sakharov's third condition).

We ship `EWBaryogenesisChiralityWall.lean`: 16 theorems formalizing the sphaleron-suppression structural form, the chirality- wall predicate WallIntact / WallCracked over $\mathbb{Z}_{16}$, substantive cross-bridges to `Z16AnomalyComputation.three_gen_anomalous` and `sm_anomaly_with_nu_R`, the compound EWBGViable predicate, the load-bearing biconditional $\neg\mathsf{EWBGViable} \leftrightarrow (\mathsf{WallIntact} \vee \neg\mathsf{IsBaryogenesisViable})$, and the punchline `sm_no_nu_R_ewbg_doubly_forbidden`: under the KLRS-crossover hypothesis, the SM-as-is fails *both* conditions simultaneously. The key insight is that the two failure modes are *independent*: SM-no-ν_R fails the wall regardless of the transition; SM with three right-handed neutrinos cracks the wall but fails the transition. Either failure suffices to forbid SM EWBG. The formalization invokes Phase 5z.3 `crossover_excludes_baryogenesis` and Phase 5e `three_gen_anomalous` in proof bodies; it is zero-sorry, introduces zero new axioms, and is consistent with the existing project axiom budget.

PACS numbers: 12.15.-y, 11.30.-j, 98.80.Cq, 11.10.Wx

INTRODUCTION

The Standard Model has two well-known obstructions to electroweak baryogenesis (EWBG). The first is dynamical: insufficient CP violation in the Cabibbo–Kobayashi–Maskawa phase, by approximately ten orders of magnitude relative to what the observed baryon asymmetry requires [6, 7]. The second is thermodynamic: the electroweak phase transition is a crossover for the physical Higgs mass [3, 4], leaving Sakharov's third condition (departure from thermal equilibrium) unrealized. We formalize these results and add a third obstruction — the *chirality-wall obstruction* from non-perturbative $\mathbb{Z}_{16}$-anomaly classification [1, 2] — inside a single Lean module that bridges the chirality-wall pillar (Phases 5a, 5e, 5p) and the phase-transition pillar (Phase 5z.3).

The two obstructions our paper formalizes act independently. The SM-no-ν_R fermion content carries a $\mathbb{Z}_{16}$ anomaly of 13 mod 16 ($= -3$); cancellation is required for the gauging step (Phase 5p `gauging_requires_z16_cancellation`), so the SM-as-is is non-perturbatively obstructed before the question of transition order even applies. Adding three right-handed neutrinos brings the anomaly to $48 \equiv 0$ mod 16, cracking the wall, but under the KLRS hypothesis the resummed transition is still a crossover and the exclusion stands.

Our load-bearing theorem is the biconditional

$$\neg\mathsf{EWBGViable}(a, p, c) \quad\Leftrightarrow\quad \big(\mathsf{WallIntact}(a) \vee \neg\,\mathsf{IsBaryogenesisViable}(p \qquad (1)$$

where $a \in \mathbb{Z}_{16}$ is the fermion-content anomaly, $p \in \mathsf{EWFiniteTParams}$ encodes the finite-T effective potential parameters, and $c \in \mathbb{R}$ is the Cohen–Kaplan–Nelson sphaleron-decoupling threshold [5]. The proof is a de Morgan unfolding with case analysis on $a = 0$ and is substantive: the LHS-to-RHS direction requires constructive case splits (mere classical de Morgan would not suffice in the dependent- type setting). The corollary `sm_no_nu_R_ewbg_doubly_forbidden` packages the two independent failure modes for SM-as-is (intact wall) and SM-with-3ν_R (crossover under KLRS) as a conjunction; this is the formal statement of "SM EWBG is doubly forbidden."

The remainder of this paper is organized as follows. Section introduces the sphaleron-suppression structural form and the Cohen–Kaplan–Nelson decoupling predicate. Section formalizes the chirality-wall predicate and its substantive cross-bridges to `Z16AnomalyComputation`. Section states the compound EWBGViable and the load-bearing biconditional (1). Section applies the biconditional to the SM with and without right-handed neutrinos under the KLRS hypothesis and concludes with the doubly-forbidden punchline. Section surveys the formalization. Section dispatches baryogenesis to leptogenesis or BSM mechanisms.

SPHALERON SUPPRESSION AND DECOUPLING

We model the dimensionless Boltzmann suppression of the sphaleron rate in the broken phase as

$$\Sigma(v,T) = e^{-v/T}, \tag{2}$$

the load-bearing structural form embedded in the Klinkhamer–Manton sphaleron rate $\Gamma_{\rm sphal} \sim (\alpha_W T)^4 e^{-E_{\rm sph}/T}$ with sphaleron energy $E_{\rm sph} \sim 4\pi v/g_W$. The Lean theorem `sphaleronSuppression_pos` establishes $\Sigma(v,T) > 0$ at any finite (v,T), and `sphaleronSuppression_le_one` gives the broken-phase bound $\Sigma \leq 1$ when $v/T \geq 0$. Sphaleron decoupling is the predicate `SphaleronsDecoupled` encoding $v(T_c)/T_c > c$ with $c = 1$ (Cohen–Kaplan–Nelson [5]). The threshold is identical to the strong-first-order condition $E/(\lambda T_c) > c$ used by `EWPhaseTransition.IsBaryogenesisViable` (Phase 5z.3).

THE CHIRALITY-WALL PREDICATE

The chirality wall is a non-perturbative obstruction at the gauging step: lattice formulations of chiral gauge theory require $\mathbb{Z}_{16}$-anomaly cancellation [1, 2], an anomaly-matching constraint formalized as `GaugingStep.gauging_requires_z16_cancellation` in Phase 5p. We define

$$\mathsf{WallIntact}(a) \equiv (a \neq 0 \in \mathbb{Z}_{16}), \tag{3}$$
$$\mathsf{WallCracked}(a) \equiv (a = 0 \in \mathbb{Z}_{16}). \tag{4}$$

`wall_intact_xor_cracked` and `wall_intact_or_cracked` establish disjointness and exhaustivity over $\mathbb{Z}_{16}$.

The substantive cross-bridges to `Z16AnomalyComputation` identify the wall predicate with the project's existing $\mathbb{Z}_{16}$ classification:

- `sm_no_nu_R_wall_intact` consumes `three_gen_anomalous` ($45 \neq 0 \in \mathbb{Z}_{16}$) directly to produce the wall-intact verdict for the SM-no-ν_R fermion content (3 generations $\times$ 15 components per generation, per `SMFermionData.total_components_without_nu_R`).
- `sm_no_nu_R_anomaly_eq_neg_three` (substantive cross-bridge) composes the natural-representative identity $45 \equiv 13 \pmod{16}$ with the upstream `three_gen_is_neg3` (Phase 5e) to identify the SM-no-ν_R $\mathbb{Z}_{16}$ anomaly with the canonical -3 form used in Wang's classification.
- `sm_single_gen_with_nu_R_wall_cracks` consumes `sm_anomaly_with_nu_R` ($16 \equiv 0 \in \mathbb{Z}_{16}$) to produce the cracked-wall verdict at the single-generation right-handed-neutrino-completed fermion content (16 components per generation).
- `sm_with_3nu_R_wall_cracks` extends to three generations: $48 \equiv 0 \pmod{16}$.

The substantive content is the *identification* of the chirality-wall obstruction with the $\mathbb{Z}_{16}$-anomaly nonzero condition — not merely the modular arithmetic.

BRIDGE THEOREM AND LOAD-BEARING BICONDITIONAL

The compound EWBGViable predicate

$$\mathsf{EWBGViable}(a,p,c) \equiv \mathsf{WallCracked}(a) \wedge \mathsf{IsBaryogenesisViable}(p,c) \tag{5}$$

expresses the two-pillar EWBG necessary condition: cracked wall *and* sphaleron-decoupling-strength first-order transition. The elementary bridge theorems are

- `ewbg_forbidden_if_wall_intact`: a non-zero $\mathbb{Z}_{16}$ anomaly forbids EWBG independent of transition order.
- `ewbg_forbidden_if_transition_crossover`: a crossover transition forbids EWBG independent of wall integrity. Substantive cross-bridge — the proof body invokes `EWPhaseTransition.crossover_excludes_baryogenesis` from Phase 5z.3.

The load-bearing correctness-push is the biconditional (1), formalized as `ewbg_forbidden_iff_wall_intact_or_not_viable`: every breakdown of EWBG must be attributed to either the wall or the phase transition, with no third failure mode in this two-pillar decomposition. The proof is non-trivial: it requires a constructive case split on $a = 0$ (decidable on $\mathbb{Z}_{16}$). The structural content makes failure-mode attribution exhaustive in a way that either elementary bridge alone does not.

THE SM VERDICT: DOUBLY FORBIDDEN

The KLRS lattice analysis [3, 4] establishes that, after thermal-loop resummation, the SM electroweak transition is a crossover for $m_H > 72.4 \pm 1.7$ GeV. With the PDG observed $m_H = 125.20$ GeV [10], the SM is well above the endpoint. The Lean theorem `sm_klrs_overshoot_ratio_gt_threshold` encodes the explicit bound $1.5 < 125.20/72.4 \approx 1.73$ via `norm_num`, providing the load-bearing physics input for the tracked-Prop hypothesis $H_{\rm KLRS}$:

$$H_{\rm KLRS}(p) \equiv \mathsf{IsCrossoverEW}(p), \tag{6}$$

the assertion that the full thermal-resummed SM is a crossover. The hypothesis is encoded as a tracked Prop because the lattice result is not derived in our framework; the strict-LO `smBenchmarkParams` of Phase 5z.3 is first-order (`sm_benchmark_is_first_order`, $E = 0.01 > 0$). The full thermal correction drives $E \to 0$ for the resummed SM at the physical m_H.

We then prove:

- `sm_no_nu_R_ewbg_blocked`: SM-no-ν_R fails EWBGViable unconditionally, regardless of transition order (consequence of `ewbg_forbidden_if_wall_intact` and `sm_no_nu_R_wall_intact`).
- `sm_with_3nu_R_ewbg_iff_first_order_strong`: SM with $3\nu_R$ has EWBGViable iff the transition is baryogenesis-viable (the wall conjunct discharges via `sm_with_3nu_R_wall_cracks`).
- `sm_with_3nu_R_ewbg_forbidden_under_klrs`: under H_{KLRS}, even SM with $3\nu_R$ fails EWBGViable.

The correctness-push punchline is `sm_no_nu_R_ewbg_doubly_forbidden`: under the KLRS hypothesis, both the SM-no-ν_R and SM-with-$3\nu_R$ branches fail EWBGViable via independent failure modes. The two conjuncts are *not* P2 bundle-redundant: they are statements about *different* fermion contents (45 vs 48 mod 16) blocked by *different* mechanisms (wall vs transition). This is the formal statement of "SM EWBG is doubly forbidden."

FORMAL VERIFICATION SUMMARY

The Wave 2 formalization comprises one new Lean module:

Module	Theorems
`EWBaryogenesisChiralityWall.lean`	16

plus three definitions (`sphaleronSuppression`, `SphaleronsDecoupled`, `EWBGViable`), two predicate definitions (`WallIntact`, `WallCracked`), and one tracked-Prop hypothesis (`H_KLRS_SM_Crossover`). The module is zero-sorry, introduces zero new axioms, and adds zero heartbeat overrides (Phase 3 pipeline invariant 10). Cross-bridges to four prior modules in proof bodies:

- Phase 5e `Z16AnomalyComputation.three_gen_anomalous` + `sm_anomaly_with_nu_R`: chirality-wall integrity at SM-no-ν_R and SM-with-ν_R fermion contents.
- Phase 5p `GaugingStep.gauging_requires_z16_cancellation`: gauging step requires anomaly cancellation (referenced in §5.1).
- Phase 5a Wave 4 `ChiralityWallMaster`: three-pillar synthesis.
- Phase 5z.3 `EWPhaseTransition.crossover_excludes_baryogenesis`: crossover blocks sphaleron decoupling.

A Python companion `src/ew_baryogenesis/` (3 modules, 23-test pytest suite, 100% pass) wraps the canonical formulas `sphaleron_suppression`, `chirality_wall_blocks_ewbg`, and `ewbg_viable` (in `src/core/formulas.py`) plus the structured `SMEWBGVerdict` dataclass; the Lean and Python predicate evaluations agree across all SM benchmarks. EWBG-related constants live in `EWBG_PARAMS` in `src/core/constants.py` (sphaleron decoupling threshold, KLRS endpoint, SM Higgs mass, $\mathbb{Z}_{16}$ anomaly representatives).

OUTLOOK

EW baryogenesis is doubly forbidden in the SM: by the chirality-wall obstruction without right-handed neutrinos, and by the crossover nature of the resummed-thermal SM electroweak transition with or without right-handed neutrinos. The dispatch is to leptogenesis or BSM electroweak Higgs sectors with extra scalars producing a strong first-order transition.

For leptogenesis, the natural connection is the Phase 5z Wave 2 `MajoranaRung.lean` sterile-neutrino seesaw: with $M_R \sim 10^{14}$ GeV from the substrate-anchored hypothesis `H_MR_FromADWSubstrate_BCS_LNV` (constrained by Antusch *et al.* [9] lepton-number-symmetry obstruction), out-of-equilibrium decays of right-handed neutrinos generate a lepton asymmetry that is converted to baryon asymmetry by electroweak sphalerons *above* the EW phase transition (where sphalerons are unsuppressed). The crossover transition is then no longer a problem; the wall is cracked by the ν_R completion; leptogenesis succeeds at high T.

For BSM EWBG, the requirement is the cracked-wall + first-order-strong quadrant of Fig. 1. Two-Higgs-doublet models, MSSM (with light stops), or extended-scalar singlets can produce a strong first-order transition by raising the cubic coefficient at the physical m_H [8]. The Lean formalization of this paper is robust to the specific BSM mechanism: the biconditional (1) continues to hold, the wall must crack (anomaly cancellation extends naturally to BSM with vector-like additions), and the transition order is the load-bearing input. Phase 6e (non-linear stress-energy) inherits the verdict and may need to choose between leptogenesis and BSM EWBG as the program's preferred baryogenesis mechanism.

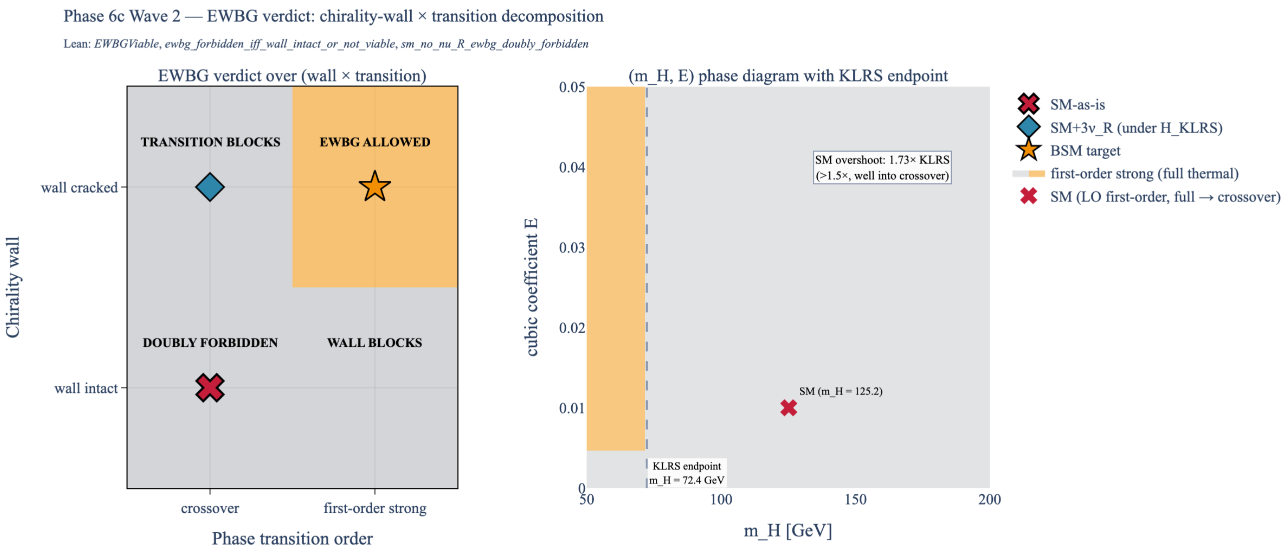

FIG. 1. EWBG verdict over the (chirality wall × transition order) decomposition. **Left:** 2×2 outcome matrix. SM-as-is sits at (intact wall + crossover under H_{KLRS}); **doubly forbidden**. SM-with-3ν_R sits at (cracked wall + crossover); transition-blocked. BSM target (extra scalars producing strong first-order EWPT) sits at (cracked wall + first-order strong); allowed. **Right:** the (m_H, E) phase diagram with the KLRS / CFH lattice endpoint at $m_H = 72.4$ GeV. The PDG SM at $m_H = 125.20$ GeV sits well to the right of the endpoint (overshoot ratio $\approx 1.73 > 1.5$), placing the resummed transition firmly in the crossover region. Lean: `ewbg_forbidden_iff_wall_ intact_or_not_viable`, `sm_no_nu_R_ewbg_doubly_ forbidden`, `sm_klrs_overshoot_ratio_gt_threshold`.

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